

## PODGORICA GRAD, Montenegro

WMO: 134630

Lat: 42.433N

Lon: 19.283E

Elev: 164

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 26.7       | 29.5 | 8.3                           | 8.5 | 35.6 | 12.4 | 10.4 | 37.5 | 29.9                  | 39.0 | 27.1 | 38.0 | 7.1                   | 0    | 0.420 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.0                   | 98.6            | 71.4 | 95.6 | 70.7 | 93.2 | 70.2 | 74.2                | 90.8 | 72.9 | 89.8 | 71.7 | 88.3 | 6.8                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 69.6                            | 109.4 | 80.7 | 67.9 | 103.2 | 79.0 | 66.3 | 97.5 | 78.0 | 37.8          | 90.8 | 36.6 | 89.9 | 35.5 | 88.3 | 85.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 23.2              | 19.3 | 15.9 | DB                         | 22.1 | 102.3              | 3.0 | 3.4                                                | 19.9 | 104.7      | 18.1 | 106.7      | 16.4 | 108.5      | 14.2 | 111.0 |
|                   |      |      | WB                         | 19.8 | 77.2               | 3.3 | 2.6                                                | 17.4 | 79.1       | 15.5 | 80.6       | 13.7 | 82.1       | 11.3 | 84.0  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 61.3   | 42.6  | 45.2  | 51.5  | 59.1  | 67.2  | 75.7  | 81.2  | 81.1  | 71.3  | 62.2  | 52.6  | 44.8  |
|                                                                        | DBStd       | 14.67  | 5.66  | 6.02  | 5.89  | 5.66  | 5.52  | 6.26  | 4.93  | 5.25  | 5.68  | 5.62  | 6.12  | 5.78  |
|                                                                        | HDD50       | 667    | 235   | 151   | 52    | 5     | 0     | 0     | 0     | 0     | 0     | 1     | 43    | 180   |
|                                                                        | HDD65       | 3029   | 695   | 553   | 418   | 189   | 41    | 4     | 0     | 0     | 0     | 119   | 372   | 628   |
|                                                                        | CDD50       | 4790   | 5     | 18    | 99    | 277   | 533   | 772   | 966   | 964   | 638   | 379   | 121   | 18    |
|                                                                        | CDD65       | 1677   | 0     | 0     | 0     | 11    | 109   | 326   | 502   | 499   | 198   | 32    | 0     | 0     |
|                                                                        | CDH74       | 18231  | 0     | 1     | 4     | 126   | 913   | 3448  | 6058  | 5884  | 1597  | 199   | 1     | 0     |
|                                                                        | CDH80       | 8573   | 0     | 0     | 0     | 21    | 229   | 1520  | 3152  | 3083  | 549   | 19    | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 5.6    | 5.5   | 6.1   | 6.8   | 5.5   | 5.6   | 5.6   | 6.3   | 5.8   | 5.4   | 4.7   | 4.8   | 5.6   |
| (15) Precipitation                                                     | PrecAvg     | 67.2   | 6.7   | 6.2   | 5.6   | 5.8   | 3.5   | 2.2   | 1.8   | 2.6   | 5.3   | 7.0   | 9.5   | 8.5   |
|                                                                        | PrecMax     | 92.9   | 15.0  | 15.2  | 13.7  | 13.4  | 9.1   | 6.4   | 5.5   | 10.8  | 15.4  | 20.6  | 25.2  | 16.3  |
|                                                                        | PrecMin     | 38.9   | 0.0   | 0.0   | 0.2   | 0.6   | 0.0   | 0.3   | 0.0   | 0.2   | 0.0   | 0.0   | 0.8   | 1.4   |
|                                                                        | PrecStd     | 12.2   | 4.5   | 3.9   | 3.2   | 3.0   | 2.2   | 1.6   | 1.2   | 2.4   | 4.1   | 4.9   | 4.8   | 4.2   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 59.9  | 65.9  | 73.5  | 82.7  | 88.2  | 97.5  | 101.5 | 102.6 | 94.7  | 82.5  | 71.9  | 62.3  |
|                                                                        |             | MCWB   | 51.6  | 52.8  | 55.8  | 62.4  | 67.2  | 72.3  | 71.7  | 71.9  | 69.1  | 65.5  | 60.3  | 55.4  |
|                                                                        | 2%          | DB     | 57.0  | 61.1  | 68.7  | 77.2  | 85.6  | 94.6  | 98.5  | 99.4  | 89.6  | 78.9  | 68.1  | 59.2  |
|                                                                        |             | MCWB   | 50.4  | 51.1  | 54.4  | 59.7  | 66.4  | 71.9  | 71.6  | 70.9  | 67.5  | 64.0  | 58.6  | 53.8  |
|                                                                        | 5%          | DB     | 53.9  | 58.6  | 65.9  | 73.6  | 82.4  | 91.4  | 95.4  | 96.4  | 86.0  | 75.6  | 64.9  | 57.0  |
|                                                                        |             | MCWB   | 48.1  | 49.9  | 53.3  | 58.1  | 65.2  | 70.6  | 70.9  | 70.3  | 67.2  | 62.7  | 57.3  | 51.7  |
|                                                                        | 10%         | DB     | 51.7  | 55.4  | 62.5  | 70.2  | 79.1  | 88.1  | 93.1  | 93.3  | 82.6  | 73.0  | 62.4  | 54.0  |
|                                                                        |             | MCWB   | 46.7  | 47.8  | 51.6  | 56.9  | 63.8  | 69.4  | 70.2  | 69.9  | 66.1  | 61.6  | 56.2  | 49.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 54.9  | 56.2  | 60.7  | 64.2  | 69.9  | 75.7  | 76.4  | 76.3  | 72.1  | 69.0  | 63.3  | 58.1  |
|                                                                        |             | MCDB   | 58.0  | 60.8  | 65.9  | 78.8  | 84.1  | 91.4  | 91.6  | 91.3  | 83.9  | 77.1  | 67.8  | 60.0  |
|                                                                        | 2%          | WB     | 52.2  | 53.5  | 57.0  | 61.9  | 67.7  | 73.5  | 74.0  | 73.9  | 70.5  | 66.8  | 61.0  | 55.6  |
|                                                                        |             | MCDB   | 55.2  | 58.9  | 64.4  | 74.0  | 81.7  | 90.5  | 91.0  | 90.5  | 82.7  | 74.3  | 65.9  | 57.9  |
|                                                                        | 5%          | WB     | 50.2  | 51.6  | 55.0  | 60.0  | 66.3  | 72.0  | 72.8  | 72.6  | 69.2  | 65.0  | 59.2  | 53.3  |
|                                                                        |             | MCDB   | 52.2  | 56.3  | 62.3  | 70.0  | 79.6  | 88.2  | 90.4  | 89.5  | 81.3  | 71.9  | 63.6  | 55.8  |
|                                                                        | 10%         | WB     | 48.2  | 49.9  | 53.3  | 58.5  | 64.8  | 70.3  | 71.5  | 71.3  | 67.8  | 63.4  | 57.2  | 50.7  |
|                                                                        |             | MCDB   | 50.2  | 53.4  | 59.8  | 67.3  | 76.9  | 85.6  | 88.9  | 88.1  | 79.2  | 69.8  | 60.9  | 53.1  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 14.8  | 15.8  | 16.7  | 18.2  | 19.0  | 19.9  | 21.0  | 21.4  | 19.1  | 18.2  | 16.0  | 14.5  |
|                                                                        |             | MCDBR  | 18.4  | 21.3  | 22.7  | 24.2  | 23.7  | 23.0  | 24.3  | 25.1  | 23.2  | 22.5  | 19.6  | 17.6  |
|                                                                        |             | MCWBR  | 12.3  | 12.9  | 12.1  | 11.2  | 10.2  | 9.1   | 8.8   | 9.2   | 9.4   | 11.1  | 11.3  | 11.6  |
|                                                                        |             | MCDBR  | 13.7  | 16.6  | 18.8  | 21.1  | 21.6  | 22.0  | 22.2  | 22.8  | 20.5  | 18.1  | 15.7  | 12.1  |
|                                                                        | 5% WB       | MCWBR  | 10.2  | 11.0  | 11.2  | 10.5  | 10.0  | 9.3   | 9.1   | 9.5   | 9.5   | 10.0  | 9.9   | 9.1   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.343 | 0.371 | 0.424 | 0.484 | 0.473 | 0.453 | 0.454 | 0.466 | 0.441 | 0.405 | 0.372 | 0.342 |
|                                                                        | taud        |        | 2.362 | 2.254 | 2.137 | 2.007 | 2.101 | 2.215 | 2.211 | 2.171 | 2.222 | 2.298 | 2.337 | 2.374 |
|                                                                        | Ebn at Noon |        | 248   | 259   | 257   | 250   | 257   | 262   | 260   | 252   | 249   | 244   | 234   | 237   |
|                                                                        | Edh at Noon |        | 26    | 34    | 43    | 53    | 50    | 44    | 44    | 45    | 40    | 33    | 27    | 24    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 513   | 741   | 1100  | 1440  | 1794  | 2085  | 2157  | 1929  | 1373  | 952   | 598   | 459   |
|                                                                        | RadStd      |        | 62    | 113   | 157   | 164   | 150   | 147   | 128   | 134   | 117   | 103   | 65    | 88    |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | +1.04   | N/A    | -2.45   | N/A   | N/A   | N/A         | -254  | N/A   | N/A   |
| (47) | Regional (27 neighbors) | +0.83   | N/A    | N/A     | N/A   | N/A   | N/A         | -185  | N/A   | N/A   |

Nomenclature: See separate page